

B6 Condensed-Matter Physics — Model Solutions, 2019

Question 1 — Reciprocal lattice and powder diffraction

(a) Lattice and basis

Problem. Define the terms *lattice* and *basis* as they have been used in this course.

Lattice: an infinite set of points $\{\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3 : n_i \in \mathbb{Z}\}$ such that the environment seen from every point is identical (a Bravais lattice).

Basis: the group of one or more atoms (with positions $\mathbf{r}_p$ relative to a lattice point) that is repeated on every lattice site to build the crystal.

(b) Reciprocal lattice

Problem. Given primitive lattice vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$, define the reciprocal lattice and prove the standard primitive vectors generate every $\mathbf{G}$ satisfying $\exp(i\mathbf{G} \cdot \mathbf{R}) = 1$.

Define

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}. \quad (1)$$

Orthogonality identity:

$$\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi \delta_{ij}.$$

Take $\mathbf{G} = m_1\mathbf{b}_1 + m_2\mathbf{b}_2 + m_3\mathbf{b}_3$ with $m_i \in \mathbb{Z}$ and $\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$:

$$\mathbf{G} \cdot \mathbf{R} = \sum_{i,j} m_i n_j (\mathbf{b}_i \cdot \mathbf{a}_j).$$

Apply orthogonality $\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi \delta_{ij}$:

$$\mathbf{G} \cdot \mathbf{R} = 2\pi \sum_i m_i n_i.$$

Sum of integers is integer:

$$\mathbf{G} \cdot \mathbf{R} \in 2\pi\mathbb{Z}.$$

Hence $e^{i\mathbf{G} \cdot \mathbf{R}} = 1$: every integer combination of $\mathbf{b}_i$ is a reciprocal lattice vector.

Conversely let $\mathbf{G} = \sum_i x_i \mathbf{b}_i$ with real x_i satisfy $e^{i\mathbf{G} \cdot \mathbf{R}} = 1$ for every lattice $\mathbf{R}$. Take $\mathbf{R} = \mathbf{a}_j$:

$$\mathbf{G} \cdot \mathbf{a}_j = \sum_i x_i (\mathbf{b}_i \cdot \mathbf{a}_j).$$

Apply orthogonality:

$$\mathbf{G} \cdot \mathbf{a}_j = 2\pi x_j.$$

Impose $e^{i\mathbf{G} \cdot \mathbf{a}_j} = e^{2\pi i x_j} = 1$:

$$x_j \in \mathbb{Z}.$$

Every $\mathbf{G}$ is an integer combination of $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$.

(c) Periodic Fourier transform

Problem. Show that $V_{\mathbf{q}} = \int d^3\mathbf{x} e^{i\mathbf{q}\cdot\mathbf{x}} V(\mathbf{x})$ vanishes unless $\mathbf{q}$ is a reciprocal lattice vector.

Shift integration variable $\mathbf{x} \rightarrow \mathbf{x} + \mathbf{R}$ for any lattice $\mathbf{R}$:

$$V_{\mathbf{q}} = \int d^3\mathbf{x} e^{i\mathbf{q}\cdot(\mathbf{x}+\mathbf{R})} V(\mathbf{x} + \mathbf{R}).$$

Use periodicity $V(\mathbf{x} + \mathbf{R}) = V(\mathbf{x})$:

$$V_{\mathbf{q}} = e^{i\mathbf{q}\cdot\mathbf{R}} \int d^3\mathbf{x} e^{i\mathbf{q}\cdot\mathbf{x}} V(\mathbf{x}).$$

Identify the remaining integral as $V_{\mathbf{q}}$:

$$V_{\mathbf{q}} = e^{i\mathbf{q}\cdot\mathbf{R}} V_{\mathbf{q}}.$$

Rearrange:

$$(1 - e^{i\mathbf{q}\cdot\mathbf{R}}) V_{\mathbf{q}} = 0 \quad \forall \mathbf{R}.$$

$$V_{\mathbf{q}} = 0 \quad \text{unless } e^{i\mathbf{q}\cdot\mathbf{R}} = 1 \quad \forall \mathbf{R}, \text{ i.e. } \mathbf{q} = \mathbf{G}.$$

(d) Structure factor

Problem. From $V(\mathbf{x}) = \sum_j u_j(\mathbf{x} - \mathbf{x}_j)$ and $\Gamma \propto |\langle \mathbf{k}' | V | \mathbf{k} \rangle|^2$, derive $I(\mathbf{G}) \propto |S(\mathbf{G})|^2$ with $S(\mathbf{G}) = \sum_p b_p(\mathbf{G}) e^{i\mathbf{G}\cdot\mathbf{r}_p}$. Specify $b_p(\mathbf{G})$.

Plane waves $|\mathbf{k}\rangle \propto e^{i\mathbf{k}\cdot\mathbf{x}}$. Matrix element:

$$\langle \mathbf{k}' | V | \mathbf{k} \rangle \propto \int d^3\mathbf{x} e^{-i(\mathbf{k}' - \mathbf{k})\cdot\mathbf{x}} V(\mathbf{x}).$$

Identify with the Fourier component $V_{-\mathbf{G}}$, $\mathbf{G} \equiv \mathbf{k}' - \mathbf{k}$:

$$\langle \mathbf{k}' | V | \mathbf{k} \rangle \propto V_{-\mathbf{G}}.$$

Insert $V(\mathbf{x}) = \sum_j u_j(\mathbf{x} - \mathbf{x}_j)$:

$$V_{-\mathbf{G}} = \sum_j \int d^3\mathbf{x} e^{-i\mathbf{G}\cdot\mathbf{x}} u_j(\mathbf{x} - \mathbf{x}_j).$$

Change variable $\mathbf{y} = \mathbf{x} - \mathbf{x}_j$:

$$V_{-\mathbf{G}} = \sum_j e^{-i\mathbf{G}\cdot\mathbf{x}_j} \int d^3\mathbf{y} e^{-i\mathbf{G}\cdot\mathbf{y}} u_j(\mathbf{y}).$$

Define atomic form factor $\tilde{u}_j(\mathbf{G}) \equiv \int d^3\mathbf{y} e^{-i\mathbf{G}\cdot\mathbf{y}} u_j(\mathbf{y})$:

$$V_{-\mathbf{G}} = \sum_j e^{-i\mathbf{G}\cdot\mathbf{x}_j} \tilde{u}_j(\mathbf{G}).$$

Split $\mathbf{x}_j = \mathbf{R} + \mathbf{r}_p$ (lattice vector + basis position):

$$V_{-\mathbf{G}} = \sum_{\mathbf{R}} e^{-i\mathbf{G}\cdot\mathbf{R}} \sum_{p \in \text{u.c.}} e^{-i\mathbf{G}\cdot\mathbf{r}_p} \tilde{u}_p(\mathbf{G}).$$

Lattice sum gives N_{cell} when $\mathbf{G}$ is a reciprocal lattice vector, else 0:

$$V_{-\mathbf{G}} = N_{\text{cell}} \sum_{p \in \text{u.c.}} e^{-i\mathbf{G} \cdot \mathbf{r}_p} \tilde{u}_p(\mathbf{G}).$$

Square and absorb constants into $I(\mathbf{G}) \propto |V_{-\mathbf{G}}|^2$:

$$I(\mathbf{G}) \propto |S(\mathbf{G})|^2, \quad S(\mathbf{G}) = \sum_{p \in \text{u.c.}} b_p(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}_p}, \quad b_p(\mathbf{G}) = \int d^3\mathbf{y} e^{-i\mathbf{G} \cdot \mathbf{y}} u_p(\mathbf{y}).$$

(The sum over j runs over every atom in the system; the sum over p runs over the single-cell basis.)

(e) Powder pattern of silicon

Problem. *Si is fcc with two atoms per unit cell at $(0,0,0)$ and $\frac{1}{4}(1,1,1)a$. Find $|\mathbf{G}|$ for the two lowest-angle powder rings.*

Conventional cubic reciprocal vectors $\mathbf{G} = (2\pi/a)(h,k,l)$. Fcc Bravais factor restricts to h,k,l all-even or all-odd.

Basis sum with $\mathbf{r}_1 = 0$, $\mathbf{r}_2 = (a/4)(1,1,1)$:

$$S_{hkl} \propto 1 + e^{i\mathbf{G} \cdot \mathbf{r}_2} = 1 + e^{i\pi(h+k+l)/2}.$$

Vanishes when $(h+k+l)/2$ is odd:

$$S = 0 \text{ if } h+k+l \equiv 2 \pmod{4}.$$

Test allowed hkl in increasing $h^2 + k^2 + l^2$:

$$(111) : h^2 + k^2 + l^2 = 3, \quad h+k+l = 3 \text{ (odd)} \Rightarrow \text{allowed.}$$

$$(200) : h^2 + k^2 + l^2 = 4, \quad h+k+l = 2 \Rightarrow \text{forbidden.}$$

$$(220) : h^2 + k^2 + l^2 = 8, \quad h+k+l = 4 \Rightarrow \text{allowed.}$$

Magnitudes:

$$|\mathbf{G}_{111}| = \frac{2\pi}{a} \sqrt{1^2 + 1^2 + 1^2}.$$

Simplify:

$$|\mathbf{G}_{111}| = \frac{2\pi\sqrt{3}}{a}.$$

For (220):

$$|\mathbf{G}_{220}| = \frac{2\pi}{a} \sqrt{2^2 + 2^2 + 0^2} = \frac{2\pi\sqrt{8}}{a}.$$

Simplify $\sqrt{8} = 2\sqrt{2}$:

$$|\mathbf{G}_{220}| = \frac{4\pi\sqrt{2}}{a}.$$

Plug in $a = 5.43 \text{ \AA}$:

$$|\mathbf{G}_{111}| = \frac{2\pi \times 1.732}{5.43} \text{ \AA}^{-1} \approx 2.00 \text{ \AA}^{-1}.$$

$$|\mathbf{G}_{220}| = \frac{2\pi \times 2.828}{5.43} \text{ \AA}^{-1} \approx 3.27 \text{ \AA}^{-1}.$$

$$|\mathbf{G}_{111}| = \frac{2\pi\sqrt{3}}{a} \approx 2.00 \text{ \AA}^{-1}, \quad |\mathbf{G}_{220}| = \frac{4\pi\sqrt{2}}{a} \approx 3.27 \text{ \AA}^{-1}.$$

Question 2 — Einstein model and melting of argon

(a) Einstein model and heat capacity

Problem. *State the assumptions of the Einstein model and derive $C(T)$ for N atoms.*

Assumptions: each atom is an independent 3D simple harmonic oscillator, all oscillators have the same frequency ω , and oscillators are quantised ($E_n = \hbar\omega(n + \frac{1}{2})$).

Single 1D oscillator partition function:

$$Z_1 = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega(n+\frac{1}{2})} = e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} (e^{-\beta\hbar\omega})^n.$$

Sum geometric series:

$$Z_1 = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}.$$

Take logarithm:

$$\ln Z_1 = -\frac{1}{2}\beta\hbar\omega - \ln(1 - e^{-\beta\hbar\omega}).$$

Differentiate $\ln Z_1$ with respect to β :

$$-\frac{\partial \ln Z_1}{\partial \beta} = \frac{\hbar\omega}{2} - \frac{-\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}}.$$

Simplify the second term:

$$\langle \epsilon \rangle = \frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1}.$$

Total energy ($3N$ oscillators):

$$E = 3N\langle \epsilon \rangle = \frac{3N\hbar\omega}{2} + \frac{3N\hbar\omega}{e^{\beta\hbar\omega} - 1}.$$

Define $T_E \equiv \hbar\omega/k_B$, so $\beta\hbar\omega = T_E/T$:

$$E = \frac{3N\hbar\omega}{2} + \frac{3N\hbar\omega}{e^{T_E/T} - 1}.$$

Differentiate with respect to T :

$$C = \frac{\partial E}{\partial T} = 3N\hbar\omega \cdot \frac{e^{T_E/T} (T_E/T^2)}{(e^{T_E/T} - 1)^2}.$$

Use $\hbar\omega = k_B T_E$:

$$C = 3Nk_B \left(\frac{T_E}{T} \right)^2 \frac{e^{T_E/T}}{(e^{T_E/T} - 1)^2}.$$

$$C(T) = 3Nk_B \left(\frac{T_E}{T} \right)^2 \frac{e^{T_E/T}}{(e^{T_E/T} - 1)^2}.$$

(b) High- T expansion

Problem. Expand $C = A_0 + A_1/T + A_2/T^2 + \dots$ in the high- T limit; find A_0, A_1, A_2 .

Apply hint with $x = \beta\hbar\omega = T_E/T$:

$$\frac{1}{e^x - 1} = \frac{1}{x} - \frac{1}{2} + \frac{x}{12} + O(x^3).$$

Multiply by $\hbar\omega$:

$$\frac{\hbar\omega}{e^x - 1} = \frac{\hbar\omega}{x} - \frac{\hbar\omega}{2} + \frac{\hbar\omega \cdot x}{12} + \dots$$

Substitute $x = \hbar\omega/(k_B T)$:

$$\frac{\hbar\omega}{e^x - 1} = k_B T - \frac{\hbar\omega}{2} + \frac{(\hbar\omega)^2}{12k_B T} + \dots$$

Add zero-point $\hbar\omega/2$:

$$\langle \epsilon \rangle = k_B T + \frac{(\hbar\omega)^2}{12k_B T} + O(T^{-3}).$$

Multiply by $3N$:

$$E = 3Nk_B T + \frac{3N(\hbar\omega)^2}{12k_B T} + O(T^{-3}).$$

Differentiate with respect to T :

$$C = 3Nk_B - \frac{3N(\hbar\omega)^2}{12k_B T^2} + \dots$$

Simplify the prefactor:

$$C = 3Nk_B - \frac{N(\hbar\omega)^2}{4k_B T^2} + \dots$$

Substitute $(\hbar\omega)^2 = (k_B T_E)^2$:

$$C = 3Nk_B - \frac{Nk_B T_E^2}{4T^2} + \dots$$

Read off:

$A_0 = 3Nk_B, \quad A_1 = 0, \quad A_2 = -\frac{Nk_B T_E^2}{4}.$
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(c) Einstein temperature of argon

Problem. From $C = (25 - 7500/T^2 + \dots) \text{ J K}^{-1}$, find T_E .

Match A_2 (units: J K):

$$-\frac{Nk_B T_E^2}{4} = -7500 \text{ J K}.$$

Match A_0 :

$$3Nk_B = 25 \text{ J K}^{-1}.$$

Solve for Nk_B :

$$Nk_B = 8.333 \text{ J K}^{-1}.$$

Rearrange the A_2 equation for T_E^2 :

$$T_E^2 = \frac{4 \times 7500}{Nk_B} \text{ K}^2.$$

Substitute Nk_B :

$$T_E^2 = \frac{4 \times 7500}{8.333} \text{ K}^2.$$

Evaluate:

$$T_E^2 = 3600 \text{ K}^2.$$

Take square root:

$$T_E = 60 \text{ K}.$$

$$\boxed{T_E \approx 60 \text{ K}.$$

(d) Mass and nearest-neighbour distance

Problem. Argon atomic weight 40, fcc with one atom per unit cell, sample volume 21 cm^3 . Find sample mass and nearest-neighbour distance.

Number of atoms from $Nk_B = 8.333 \text{ J K}^{-1}$:

$$N = \frac{8.333}{1.381 \times 10^{-23}} \approx 6.03 \times 10^{23}.$$

This is $\approx N_A$, so the sample is one mole:

$$M = \frac{N}{N_A} \times 40 \text{ g mol}^{-1} \approx 40 \text{ g}.$$

$$\boxed{M \approx 40 \text{ g}.$$

Number density:

$$n = \frac{N}{V} = \frac{6.03 \times 10^{23}}{21 \text{ cm}^3} \approx 2.87 \times 10^{22} \text{ cm}^{-3}.$$

Convert to SI:

$$n \approx 2.87 \times 10^{28} \text{ m}^{-3}.$$

Fcc has 4 atoms per conventional cubic cell of side a :

$$n = \frac{4}{a^3}.$$

Solve for a :

$$a^3 = \frac{4}{n} = \frac{4}{2.87 \times 10^{28}} \text{ m}^3 = 1.394 \times 10^{-28} \text{ m}^3.$$

Take cube root:

$$a = 5.198 \times 10^{-10} \text{ m} \approx 0.520 \text{ nm}.$$

Nearest-neighbour distance in fcc is along the face diagonal: $d = a/\sqrt{2}$:

$$d = \frac{0.520}{\sqrt{2}} \text{ nm} \approx 0.367 \text{ nm}.$$

$$\boxed{d \approx 0.37 \text{ nm}.$$

(e) Mean square displacement at $T \gg T_E$

Problem. Derive $\langle x^2 \rangle$ as a function of T, T_E, m for $T \gg T_E$.

1D oscillator: $H = p^2/2m + \frac{1}{2}m\omega^2 x^2$. Equipartition for $T \gg T_E$:

$$\frac{1}{2}m\omega^2 \langle x^2 \rangle = \frac{1}{2}k_B T.$$

Solve for $\langle x^2 \rangle$:

$$\langle x^2 \rangle = \frac{k_B T}{m\omega^2}.$$

Substitute $\omega = k_B T_E / \hbar$:

$$\langle x^2 \rangle = \frac{k_B T}{m(k_B T_E / \hbar)^2}.$$

Simplify:

$$\boxed{\langle x^2 \rangle = \frac{\hbar^2 T}{m k_B T_E^2}.$$

(f) Lindemann melting temperature

Problem. Use $\sqrt{\langle x^2 \rangle} = 0.05d$ to estimate T_m of argon.

Square the criterion:

$$\langle x^2 \rangle = (0.05)^2 d^2 = 2.5 \times 10^{-3} d^2.$$

Set equal to result of (e):

$$\frac{\hbar^2 T_m}{m k_B T_E^2} = 2.5 \times 10^{-3} d^2.$$

Solve for T_m :

$$T_m = \frac{2.5 \times 10^{-3} m k_B T_E^2 d^2}{\hbar^2}.$$

Insert $m = 40 \times 1.66 \times 10^{-27} \text{ kg} = 6.64 \times 10^{-26} \text{ kg}$, $T_E = 60 \text{ K}$, $d = 3.67 \times 10^{-10} \text{ m}$:

$$m k_B T_E^2 = 6.64 \times 10^{-26} \times 1.381 \times 10^{-23} \times 3600 \text{ kg J K}^{-1} = 3.30 \times 10^{-45} \text{ kg J K}^{-1}.$$

$$d^2 = (3.67 \times 10^{-10})^2 \text{ m}^2 = 1.347 \times 10^{-19} \text{ m}^2.$$

$$\hbar^2 = (1.055 \times 10^{-34})^2 \text{ J}^2 \text{ s}^2 = 1.113 \times 10^{-68} \text{ J}^2 \text{ s}^2.$$

$$T_m = \frac{2.5 \times 10^{-3} \times 3.30 \times 10^{-45} \times 1.347 \times 10^{-19}}{1.113 \times 10^{-68}} \text{ K}.$$

$$T_m \approx \frac{1.111 \times 10^{-66}}{1.113 \times 10^{-68}} \text{ K} \approx 100 \text{ K}.$$

$$\boxed{T_m \approx 100 \text{ K}.$$

(Experimental melting point of argon is 84 K at 1 atm — order of magnitude correct.)

Question 3 — 2D semiconductor quantum well (InAs)

(a) Electron density in 2D conduction band

Problem. Show $n = C(m_e^*)^a T^b e^{-\beta(E_c - \mu)}$ and find C, a, b .

2D parabolic band: $E(k) = E_c + \hbar^2 k^2 / (2m_e^*)$. Density of states per unit area (including spin $\times 2$):

$$g(E) = \frac{m_e^*}{\pi \hbar^2} \quad \text{for } E > E_c.$$

Boltzmann occupation:

$$n = \int_{E_c}^{\infty} dE g(E) e^{-\beta(E - \mu)}.$$

Pull out μ , substitute $\epsilon = E - E_c$:

$$n = \frac{m_e^*}{\pi \hbar^2} e^{-\beta(E_c - \mu)} \int_0^{\infty} d\epsilon e^{-\beta\epsilon}.$$

Evaluate exponential integral:

$$\int_0^{\infty} e^{-\beta\epsilon} d\epsilon = \frac{1}{\beta} = k_B T.$$

Combine:

$$n = \frac{m_e^* k_B}{\pi \hbar^2} T e^{-\beta(E_c - \mu)}, \quad C = \frac{k_B}{\pi \hbar^2}, \quad a = 1, \quad b = 1.$$

(b) Hole density and np

Problem. Give p and np in 2D.

By the same derivation with $E_v - E$:

$$p = \frac{m_h^* k_B}{\pi \hbar^2} T e^{-\beta(\mu - E_v)}.$$

Product (intrinsic):

$$np = \left(\frac{k_B T}{\pi \hbar^2} \right)^2 m_e^* m_h^* e^{-\beta E_g}, \quad E_g \equiv E_c - E_v.$$

(c) Intrinsic density at 300 K and intrinsic–extrinsic crossover

Problem. For InAs at 300 K: find n_i , then estimate T at which doping ($n_d = 10^9 \text{ cm}^{-2}$) changes regime.

Intrinsic: $n = p = n_i = \sqrt{np}$:

$$n_i = \frac{k_B T}{\pi \hbar^2} \sqrt{m_e^* m_h^*} e^{-E_g / (2k_B T)}.$$

At $T = 300 \text{ K}$, $k_B T = 0.02585 \text{ eV} = 4.142 \times 10^{-21} \text{ J}$:

$$\pi \hbar^2 = \pi \times (1.055 \times 10^{-34})^2 \text{ J}^2 \text{ s}^2 = 3.497 \times 10^{-68} \text{ J}^2 \text{ s}^2.$$

Divide:

$$\frac{k_B T}{\pi \hbar^2} = \frac{4.142 \times 10^{-21}}{3.497 \times 10^{-68}} \text{ m}^{-2} \text{ kg}^{-1} = 1.184 \times 10^{47} \text{ m}^{-2} \text{ kg}^{-1}.$$

Compute the effective-mass geometric mean:

$$\sqrt{m_e^* m_h^*} = \sqrt{0.026 \times 0.41} m_e.$$

Evaluate:

$$\sqrt{m_e^* m_h^*} = 0.1033 m_e = 9.40 \times 10^{-32} \text{ kg}.$$

Multiply the two factors:

$$\frac{k_B T}{\pi \hbar^2} \sqrt{m_e^* m_h^*} = 1.184 \times 10^{47} \times 9.40 \times 10^{-32} \text{ m}^{-2}.$$

Evaluate:

$$\frac{k_B T}{\pi \hbar^2} \sqrt{m_e^* m_h^*} = 1.113 \times 10^{16} \text{ m}^{-2} = 1.113 \times 10^{12} \text{ cm}^{-2}.$$

Boltzmann exponent:

$$\frac{E_g}{2k_B T} = \frac{0.36}{2 \times 0.02585} = 6.96.$$

Exponentiate:

$$e^{-6.96} = 9.49 \times 10^{-4}.$$

Combine:

$$n_i = 1.113 \times 10^{12} \times 9.49 \times 10^{-4} \text{ cm}^{-2}.$$

Evaluate:

$$n_i \approx 1.06 \times 10^9 \text{ cm}^{-2}.$$

$$\boxed{n_i(300 \text{ K}) \approx 1.1 \times 10^9 \text{ cm}^{-2}.}$$

Extrinsic–intrinsic crossover at $n_i(T^*) = n_d = 10^9 \text{ cm}^{-2}$, essentially the same condition:

$$\boxed{T^* \approx 300 \text{ K}.}$$

(Below T^* the donors dominate; above it intrinsic carriers dominate.)

(d) Donor binding in 2D

Problem. Donor wavefunction $\psi(\mathbf{r}) = A e^{-b|\mathbf{r}-\mathbf{r}_0|}$ in 2D. Find b and binding energy.

Set $\mathbf{r}_0 = 0$. Hydrogenic Hamiltonian with effective mass m_e^* and dielectric ϵ_r :

$$H = -\frac{\hbar^2}{2m_e^*} \nabla^2 - \frac{e^2}{4\pi\epsilon_0\epsilon_r r}.$$

Compute $d\psi/dr$:

$$\frac{d\psi}{dr} = -b A e^{-br} = -b\psi.$$

Apply 2D radial Laplacian:

$$\nabla^2 \psi = \frac{1}{r} \frac{d}{dr} \left(r \frac{d\psi}{dr} \right) = \frac{1}{r} \frac{d}{dr} (-br\psi).$$

Carry out the derivative:

$$\nabla^2 \psi = \frac{1}{r} (-b\psi - br(-b\psi)) = \left(b^2 - \frac{b}{r} \right) \psi.$$

Apply $H\psi = E\psi$:

$$\left[-\frac{\hbar^2}{2m_e^*} \left(b^2 - \frac{b}{r} \right) - \frac{e^2}{4\pi\epsilon_0\epsilon_r r} \right] \psi = E\psi.$$

Demand the $1/r$ coefficient vanish:

$$\frac{\hbar^2 b}{2m_e^*} - \frac{e^2}{4\pi\epsilon_0\epsilon_r} = 0.$$

Solve for b :

$$b = \frac{m_e^* e^2}{2\pi\epsilon_0\epsilon_r \hbar^2}.$$

Define effective Bohr radius $a_0^* = 4\pi\epsilon_0\epsilon_r \hbar^2 / (m_e^* e^2)$:

$$b = \frac{2}{a_0^*}.$$

Constant term gives energy:

$$E = -\frac{\hbar^2 b^2}{2m_e^*}.$$

Substitute b :

$$E = -\frac{\hbar^2}{2m_e^*} \left(\frac{m_e^* e^2}{2\pi\epsilon_0\epsilon_r \hbar^2} \right)^2.$$

Simplify:

$$E = -\frac{m_e^* e^4}{2(2\pi\epsilon_0\epsilon_r)^2 \hbar^2}.$$

Compare with effective Rydberg $\text{Ry}^* = m_e^* e^4 / [2(4\pi\epsilon_0\epsilon_r)^2 \hbar^2]$:

$$E_{2D} = -4 \text{Ry}^*.$$

Numerical Rydberg with $m_e^* = 0.026 m_e$, $\epsilon_r = 15$:

$$\text{Ry}^* = 13.6 \text{ eV} \times \frac{0.026}{15^2}.$$

Evaluate:

$$\text{Ry}^* = 13.6 \times 1.156 \times 10^{-4} \text{ eV} \approx 1.57 \times 10^{-3} \text{ eV}.$$

Multiply by 4:

$$|E_b| = 4 \times 1.57 \times 10^{-3} \text{ eV} \approx 6.3 \text{ meV}.$$

$$b = \frac{2}{a_0^*}, \quad |E_b| = 4 \text{Ry}^* \approx 6.3 \text{ meV}.$$

(e) $\log \sigma$ vs $1/T$ sketch

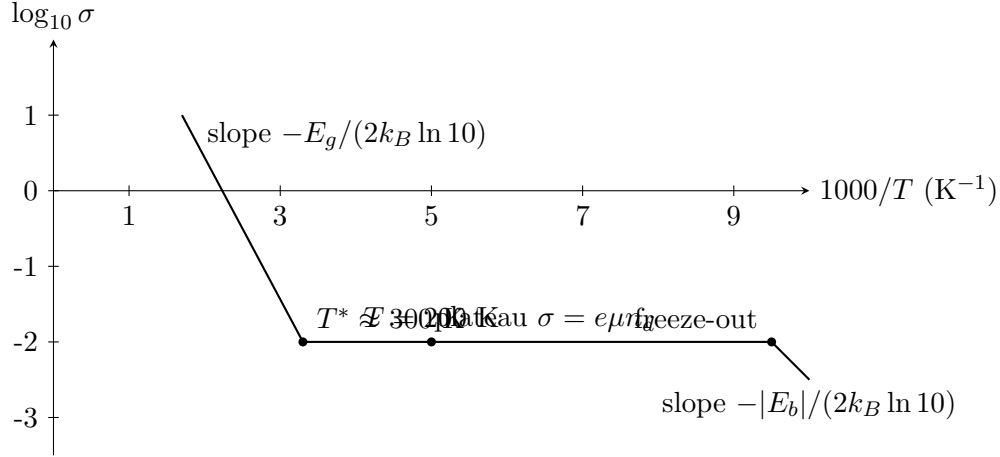
Problem. Sketch $\log \sigma$ versus $1/T$ for $30 \leq T \leq 600 \text{ K}$, given $\sigma(200 \text{ K}) = 0.01 \Omega^{-1}$, equal T -independent mobilities.

Three regimes (slopes set by activation energies $E_g/2$ for intrinsic, $|E_b|/2$ for freeze-out; flat extrinsic plateau in between):

- $T \gtrsim T^* \approx 300 \text{ K}$: intrinsic, slope $-E_g/(2k_B)$.
- $|E_b|/k_B \lesssim T \lesssim T^*$: extrinsic plateau $\sigma \approx e\mu n_d$.

- $T \lesssim |E_b|/k_B \approx 73 \text{ K}$: freeze-out, slope $-|E_b|/(2k_B)$.

Numerical anchors: at 200 K, plateau value $\sigma = 10^{-2} \Omega^{-1}$, so $\log_{10} \sigma = -2$. At 600 K ($1/T \approx 1.7 \times 10^{-3} \text{ K}^{-1}$) the intrinsic carrier density rises fast.



Steep intrinsic slope, flat extrinsic plateau at $\sigma = 10^{-2} \Omega^{-1}$, freeze-out steep again at low T .

Question 4 — Magnetism of helium

(a) Definitions

Paramagnet: substance with positive magnetic susceptibility $\chi > 0$; magnetisation aligns with applied field, vanishes when field is removed (e.g. unpaired spins).

Diamagnet: substance with negative susceptibility $\chi < 0$; induced moment opposes applied field (universal Larmor response of closed shells).

Ferromagnet: substance with spontaneous magnetisation below a Curie temperature T_c , due to cooperative alignment of moments; χ diverges at T_c .

(b) Diamagnetic susceptibility of ^4He

Problem. What magnetic behaviour does ^4He exhibit, and derive χ in terms of helium density, $\langle r^2 \rangle$, fundamental constants.

^4He is a closed-shell atom ($1s^2$, $S = L = J = 0$) with no nuclear spin: purely **diamagnetic** (Larmor).

Minimal coupling $\mathbf{p} \rightarrow \mathbf{p} + e\mathbf{A}$ with $\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}$ gives, per electron:

$$\Delta H = \frac{e^2}{8m_e}(\mathbf{B} \times \mathbf{r})^2.$$

Pick $\mathbf{B} = B\hat{z}$:

$$(\mathbf{B} \times \mathbf{r})^2 = B^2(x^2 + y^2).$$

Substitute:

$$\Delta H = \frac{e^2 B^2}{8m_e}(x^2 + y^2).$$

Spherical symmetry $\langle x^2 \rangle = \langle y^2 \rangle = \langle z^2 \rangle = \langle r^2 \rangle/3$:

$$\langle x^2 + y^2 \rangle = \frac{2}{3}\langle r^2 \rangle.$$

Multiply by 2 electrons per atom:

$$\Delta E_{\text{atom}} = \frac{e^2 B^2}{8m_e} \cdot 2 \cdot \frac{2}{3} \langle r^2 \rangle.$$

Simplify:

$$\Delta E_{\text{atom}} = \frac{e^2 B^2}{6m_e} \langle r^2 \rangle.$$

Magnetisation per atom $m_{\text{atom}} = -\partial \Delta E_{\text{atom}} / \partial B$:

$$m_{\text{atom}} = -\frac{e^2 B}{3m_e} \langle r^2 \rangle.$$

Multiply by number density n :

$$M = -\frac{ne^2}{3m_e} \langle r^2 \rangle B.$$

Use $M = \chi H$ and $B = \mu_0 H$:

$$\chi_{\text{dia}} = \frac{M}{H} = -\frac{n\mu_0 e^2}{3m_e} \langle r^2 \rangle.$$

$$\chi_{\text{dia}} = -\frac{n\mu_0 e^2}{3m_e} \langle r^2 \rangle.$$

(c) Nuclear paramagnetic susceptibility of ^3He

Problem. Independent ^3He nuclei, $\mu_{^3\text{He}} = -2.13 \mu_N$, spin $1/2$. Find $\chi_{\text{nuc}}(T, n)$.

Spin-1/2 nucleus in field B has two levels $E_{\pm} = \mp |\mu| B$:

$$Z_1 = e^{\beta |\mu| B} + e^{-\beta |\mu| B} = 2 \cosh(\beta |\mu| B).$$

Mean moment per atom $\langle m \rangle = \beta^{-1} \partial \ln Z_1 / \partial B$:

$$\langle m \rangle = |\mu| \tanh(\beta |\mu| B).$$

Expand for $|\mu| B \ll k_B T$ (linear response):

$$\langle m \rangle \approx \frac{|\mu|^2 B}{k_B T}.$$

Multiply by density n :

$$M = \frac{n |\mu|^2 B}{k_B T}.$$

Apply $\chi = M/H = \mu_0 M/B$:

$$\chi_{\text{nuc}} = \frac{n \mu_0 |\mu|^2}{k_B T}.$$

Insert $|\mu| = 2.13 \mu_N$:

$$\chi_{\text{nuc}}(T, n) = \frac{n \mu_0 (2.13 \mu_N)^2}{k_B T}.$$

(d) Sign-change temperature

Problem. Estimate T at which total χ changes sign.

Set total $\chi = \chi_{\text{dia}} + \chi_{\text{nuc}} = 0$:

$$\frac{n\mu_0(2.13\mu_N)^2}{k_B T^*} = \frac{n\mu_0 e^2}{3m_e} \langle r^2 \rangle.$$

Cancel $n\mu_0$:

$$\frac{(2.13\mu_N)^2}{k_B T^*} = \frac{e^2 \langle r^2 \rangle}{3m_e}.$$

Solve for T^* :

$$T^* = \frac{3m_e(2.13\mu_N)^2}{k_B e^2 \langle r^2 \rangle}.$$

Estimate $\langle r^2 \rangle \sim a_0^2$ with $a_0 = 0.529 \times 10^{-10} \text{ m}$:

$$\langle r^2 \rangle \approx 2.80 \times 10^{-21} \text{ m}^2.$$

Compute $(2.13\mu_N)^2$ with $\mu_N = 5.05 \times 10^{-27} \text{ J T}^{-1}$:

$$2.13\mu_N = 1.076 \times 10^{-26} \text{ J T}^{-1}.$$

Square:

$$(2.13\mu_N)^2 = 1.158 \times 10^{-52} \text{ J}^2 \text{ T}^{-2}.$$

Numerator:

$$3m_e(2.13\mu_N)^2 = 3 \times 9.11 \times 10^{-31} \times 1.158 \times 10^{-52}.$$

Evaluate:

$$3m_e(2.13\mu_N)^2 = 3.165 \times 10^{-82} \text{ kg J}^2 \text{ T}^{-2}.$$

Denominator factors:

$$e^2 = (1.602 \times 10^{-19})^2 = 2.566 \times 10^{-38} \text{ C}^2.$$

Combine:

$$k_B e^2 \langle r^2 \rangle = 1.381 \times 10^{-23} \times 2.566 \times 10^{-38} \times 2.80 \times 10^{-21}.$$

Evaluate:

$$k_B e^2 \langle r^2 \rangle = 9.92 \times 10^{-82} \text{ J C}^2 \text{ m}^2.$$

Form ratio (using $1 \text{ J/T}^2 = 1 \text{ C}^2 \text{ m}^2/\text{kg}$, so $kg \cdot J^2/T^2 = J \cdot C^2 m^2$, units of K from division by k_B):

$$T^* \approx \frac{3.165 \times 10^{-82}}{9.92 \times 10^{-82}} \text{ K}.$$

Evaluate:

$$T^* \approx 0.32 \text{ K}.$$

$T^* \sim 0.3 \text{ K (of order 100 mK–1 K)}.$

(e) Mean-field T_c for bcc ^3He

Problem. Bcc ^3He , $J_{nn}/k_B = 1 \text{ mK}$, $J_{nnn}/k_B = -0.5 \text{ mK}$, spin $1/2$. Find T_c .

Bcc lattice: $z_1 = 8$ nearest neighbours at coupling J_1 , $z_2 = 6$ next-nearest at J_2 . Effective sum:

$$J_{\text{eff}} \equiv \sum_j J_{ij} = z_1 J_1 + z_2 J_2.$$

Insert numbers:

$$J_{\text{eff}}/k_B = 8(1) + 6(-0.5) \text{ mK}.$$

Simplify:

$$J_{\text{eff}}/k_B = (8 - 3) \text{ mK} = 5 \text{ mK}.$$

Mean-field decoupling $\mathbf{S}_i \cdot \mathbf{S}_j \rightarrow \mathbf{S}_i \cdot \langle \mathbf{S}_j \rangle + \langle \mathbf{S}_i \rangle \cdot \mathbf{S}_j - \langle \mathbf{S}_i \rangle \cdot \langle \mathbf{S}_j \rangle$ on $H = -\frac{1}{2} \sum_{i,j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$ gives effective single-site Hamiltonian:

$$H_i^{\text{MF}} = -\mathbf{S}_i \cdot \mathbf{h}_{\text{eff}}, \quad \mathbf{h}_{\text{eff}} = \sum_j J_{ij} \langle \mathbf{S}_j \rangle = J_{\text{eff}} \langle \mathbf{S} \rangle.$$

High- T Brillouin self-consistency for spin S :

$$\langle S \rangle = \frac{S(S+1)}{3k_B T} h_{\text{eff}} = \frac{S(S+1)}{3k_B T} J_{\text{eff}} \langle S \rangle.$$

Non-trivial solution requires coefficient unity at T_c :

$$k_B T_c = \frac{S(S+1)}{3} J_{\text{eff}}.$$

Insert $S = 1/2$, so $S(S+1) = 3/4$:

$$k_B T_c = \frac{1}{3} \cdot \frac{3}{4} J_{\text{eff}} = \frac{1}{4} J_{\text{eff}}.$$

Evaluate:

$$T_c = \frac{1}{4} \times 5 \text{ mK} = 1.25 \text{ mK}.$$

$T_c \approx 1.25 \text{ mK}.$